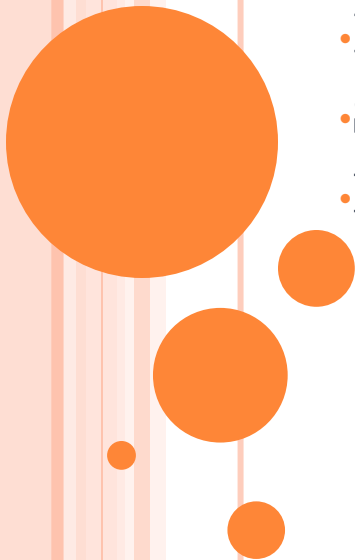


CHAPTER :1

1.1 MANAGEMENT

- **Meaning and Concept of management**
- **Function of management**
- **Scope and Application of Management**
- **Importance of Management**



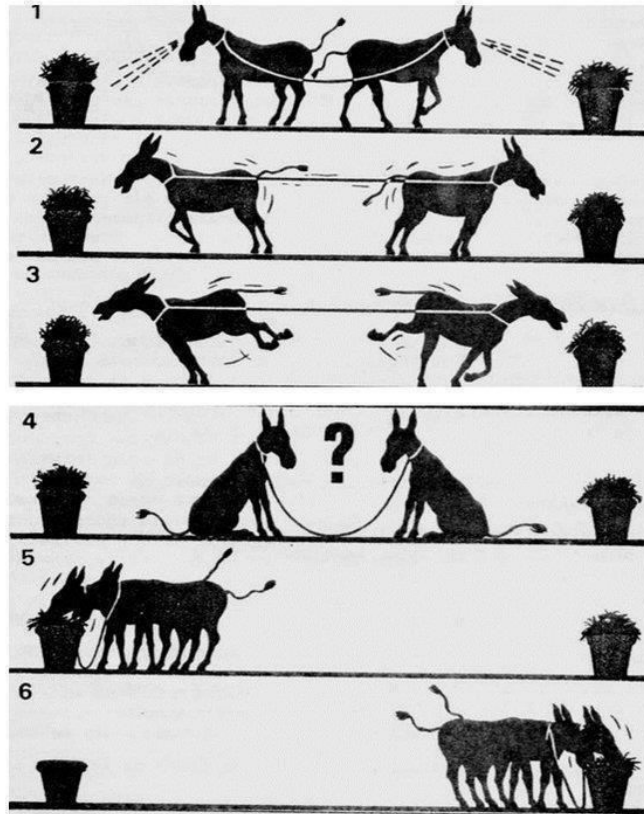
MEANING AND CONCEPT OF MANAGEMENT

- Management is the process of manages the organization. It coordinates resources to get the jobs done for achieving goals. Resources can be human, financial, physical and informational.
- *It achieves goals with **efficiency and effectiveness**.*
- *“Management is the art of getting things done through and with people in formally organized groups.” By Koontz.*
- *“Management is knowing exactly what you want people to do, and they seeing that they do it in the best and cheapest way.” By F.W. Taylor*



MANAGEMENT :—

*‘There are no **underdeveloped** countries,
there are **undermanaged** countries.’*



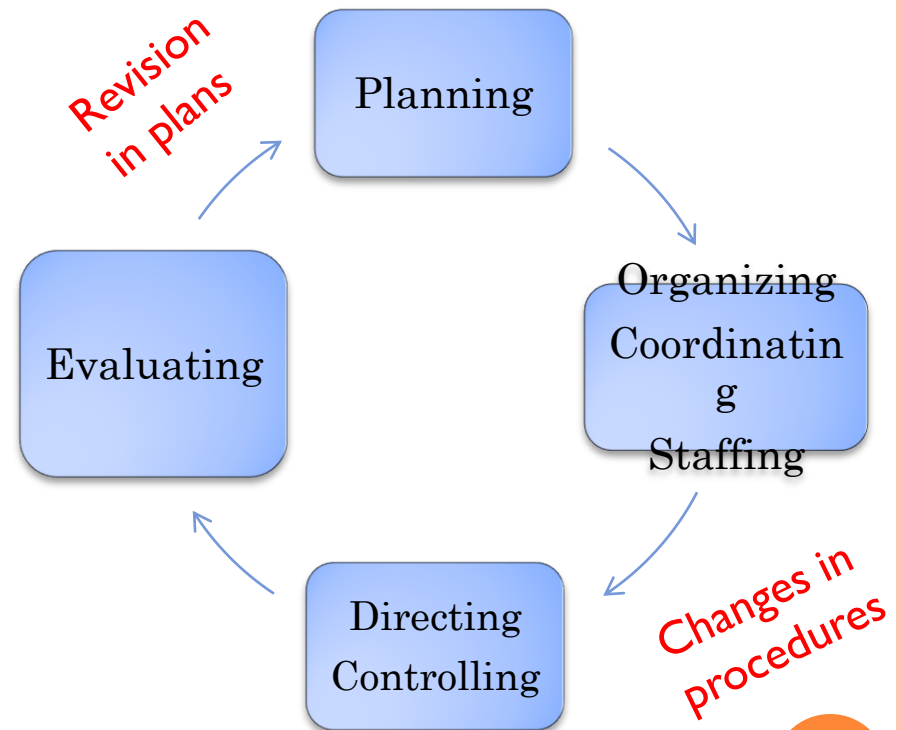
MANAGEMENT

Is management an art or science ?

ART: Because it depends on the skills, aptitude & creativity of the manager

SCIENCE: Because there is considerable knowledge in the field of management with basic principles for guidance of basic activities.

Management Process



EFFICIENCY AND EFFECTIVENESS

Efficiency

- **Efficiency** = getting the most output from the least amount of inputs
- “doing things right”
- concern with means(ways) of getting things done

Effectiveness

- **Effectiveness** = do those work activities that will help the organization reach its goals
- “doing the right things”
- concern with ends(result) of organizational goal achievement



EFFICIENCY AND EFFECTIVENESS

Efficiency

- **Efficiency** = getting work done with a minimum effort, expense, or waste
- (use resources – people, money, raw materials– wisely and cost-effectively)

Effectiveness

- **Effectiveness** = accomplish tasks that help fulfill organizational objectives
- (make the right decisions and successfully carry them out to accomplish the org. goal)



EFFICIENCY AND EFFECTIVENESS IN MANAGEMENT



Efficiency
(Means)

Effectiveness
(Ends)



**Resource
Usage**

**Goal
Attainment**



Low Waste

**High
Attainment**



**Management Strives for:
Low Resource Waste (high efficiency)
High Goal Attainment (high
effectiveness)**



FUNCTIONS OF MANAGEMENT

Management is an ongoing activity consisting of number of functions. These are:

- A. *Planning*
- B. *Organizing*
- C. *Staffing*
- D. *Directing / Leading*
- E. *Controlling*
- F. *Motivating*
- G. *Co-ordinating*
- H. *Communicating*

PLANNING

Planning is the most basic of all management functions. Planning means *thinking before doing*. In other words, planning is *the preparation for action*.

Planning includes *forecasting, formulation of objectives, policies, programs, schedules, procedures and budget, time bound*.

Strategic Planning:	Policy oriented-Long term
Tactical Planning:	Action oriented-Short term
Operational Planning:	Day to Day

ORGANIZING

Organizing involves identification and grouping the activities to be performed and dividing them among the individuals and creating authority and responsibility, Assign people to best positions, relationship among them.

Steps in Organizing

- a. Determination of activities.
- b. Division of activities.
- c. Fitting individuals into jobs.
- d. Developing relationships in terms of authorities and responsibilities.

STAFFING

Staffing involves Human Resource Management functions

Determining Job descriptions and Specifications

Recruitment and Placement of the Employees in the organization

Promotion, Remunerating and Retiring of employees in the organization

DIRECTING/LEADING

Directing involves motivating, guiding and supervising subordinates towards the organization objectives. Only giving orders is not directing. Good planning may ensure the achievement of the predetermined objectives only when the human efforts, largely diverse are co-ordinated, guided and directed for the accomplishment of the objectives.

Directing is that part of the management process which actuates the organizational members to work effectively and efficiently for the attainment of organizational objectives.

Steps in Directing

- a. Issue of orders and instructions.
- b. Guidance and training of subordinates.
- c. Supervision of subordinates' work.

CONTROLLING

It means setting standards, measuring actual performance, and taking corrective action.

Steps in Controlling

- a. Setting standards.
- b. Checking and reporting on performance.
- c. Taking corrective action.

Types of Control: based on objectives

- a. **Physical Control:** quantity and quality of products
- b. **Financial Control:** cost per unit production, cost of material, labor expenses etc.
- c. **Budgetary Control:** physical and financial standards for future are determined and results are compared against these pre-determined standards.

MOTIVATING

Motivation means inspiring people to intensify their desire and willingness to perform their duties effectively and co-operate for the achievement of common objectives of the business.

Fundamentals of Motivation

The first fundamental thing is that a person wants to exist and survive and for this he needs basic necessities of life *e.g. food, cloth, shelter, education and medical aid etc.*

The second fundamental of motivation is the desire to achieve a goal, for satisfaction or bliss. Basically, people are motivated to put in sincere efforts if they are assured of fulfilling their needs. Such as *psychological needs, social needs, security needs, ego (needs for self-respect) etc.*

MOTIVATING

Classification of Motivation

- a. **Internal Motivation:** E.g. Interests, emotional attachments, burning desires, fighting spirits for noble cause etc.
- b. **External Motivation:** E.g. Attractive salary, bonus, praise, incentive, punishment, fear of loss of job etc.

Essentials of Sound Motivation System

- a. Good wages.
- b. Good financial incentive scheme.
- c. Opportunities for achievement, growth, motivation.
- d. Human relations.
- e. Good working conditions.
- f. Job satisfaction.
- g. Job security.

CO-ORDINATING

Co-ordination may be defined as on-going process whereby a manager develops an integrated, orderly and synchronized pattern of group effort among his subordinates and tries to attain unity of effort in the pursuit of a common purpose.

Types of Co-ordination

- a. Internal Co-ordination
- b. External Co-ordination
- c. Vertical Co-ordination
- d. Horizontal Co-ordination

COMMUNICATING

Communication is the process by which instructions, ideas, thoughts or information are transmitted, received and understood, by the persons working in organization.

Components of Communication

- a. *Sender*
- b. *Message*
- c. *Receiver*
- d. *Feedback*

Effective Communication reduces minimizes the confusion among the employees in the organization which is essential for the achievement of goals

LEVELS OF MANAGEMENT

**Top Level
Management**

**Middle Level
Management**

Lower Level Management

LEVELS OF MANAGEMENT

Top level Management:

- Responsible for the performance of all departments and have cross-departmental responsibility.
- Establish organizational goals and monitor middle managers
- Decide how different departments should interact
- Ultimately responsible for the success or failure of an organization
- Includes Board of Directors, CEO, Chairman, Officer etc.

LEVELS OF MANAGEMENT

Middle level Management:

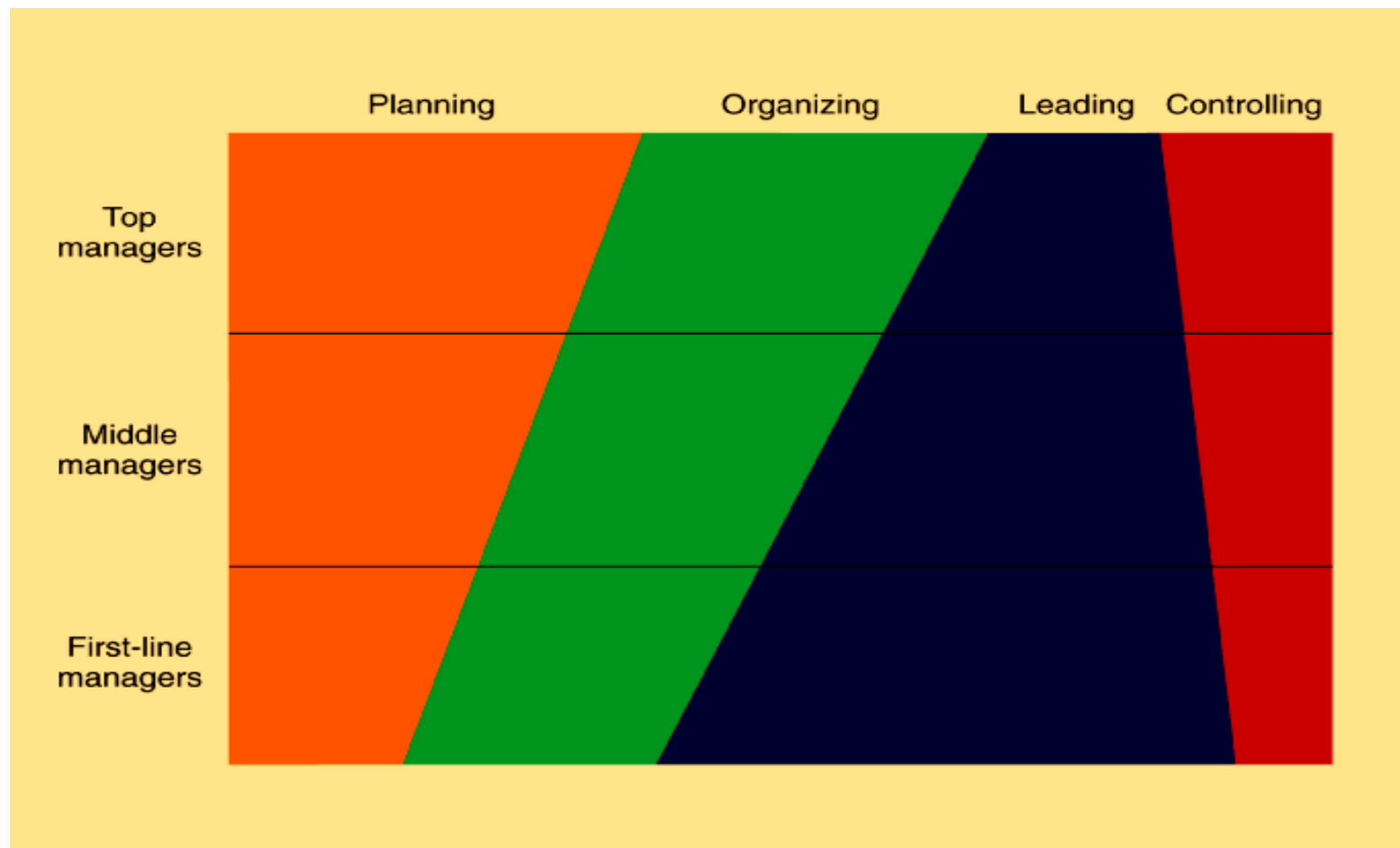
- Responsible for finding the best way to organize human and other resources to achieve organizational goals
- Establish organizational goals and monitor middle managers
- Includes Head of Department, Section officer etc.

Lower level Management:

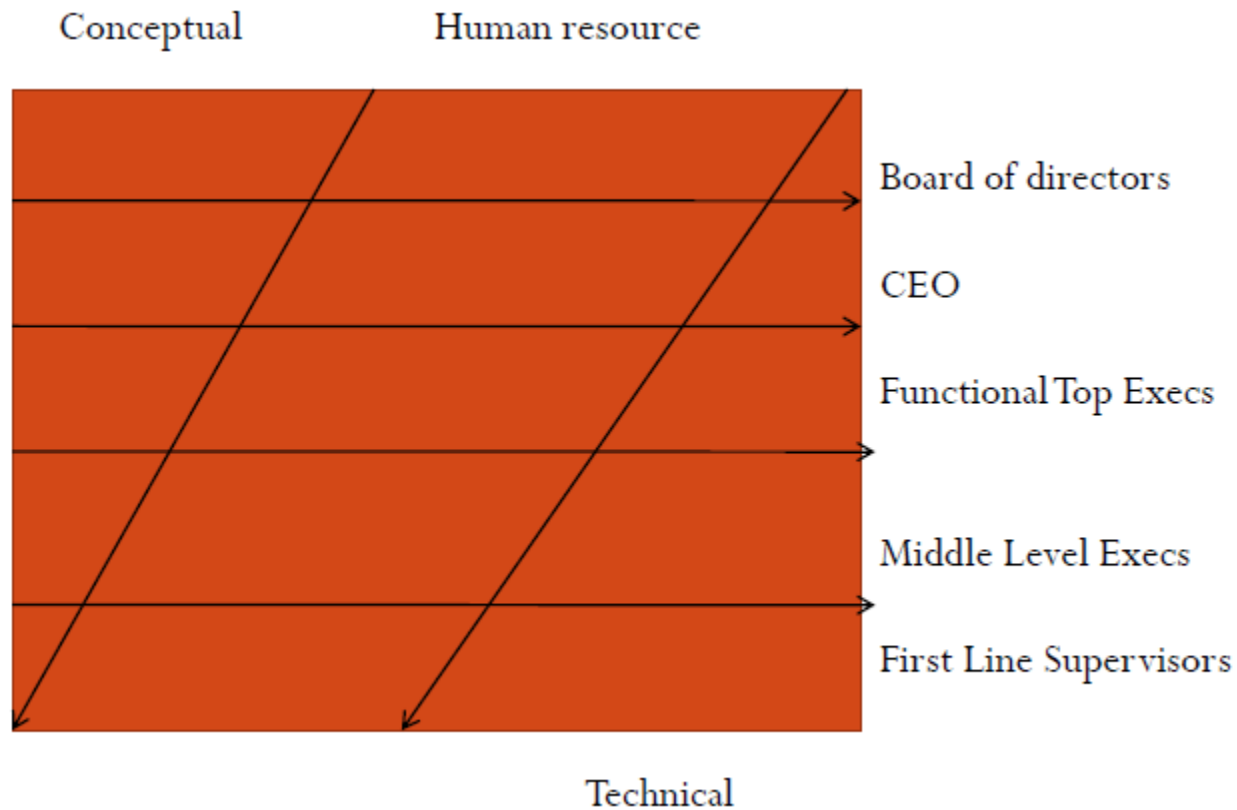
- Responsible for daily supervision of the non-managerial employees who perform many of the specific activities necessary to produce goods and services.
- Includes Foreman, Supervisor, Inspector, Superintendent, Workers

LEVELS OF MANAGEMENT

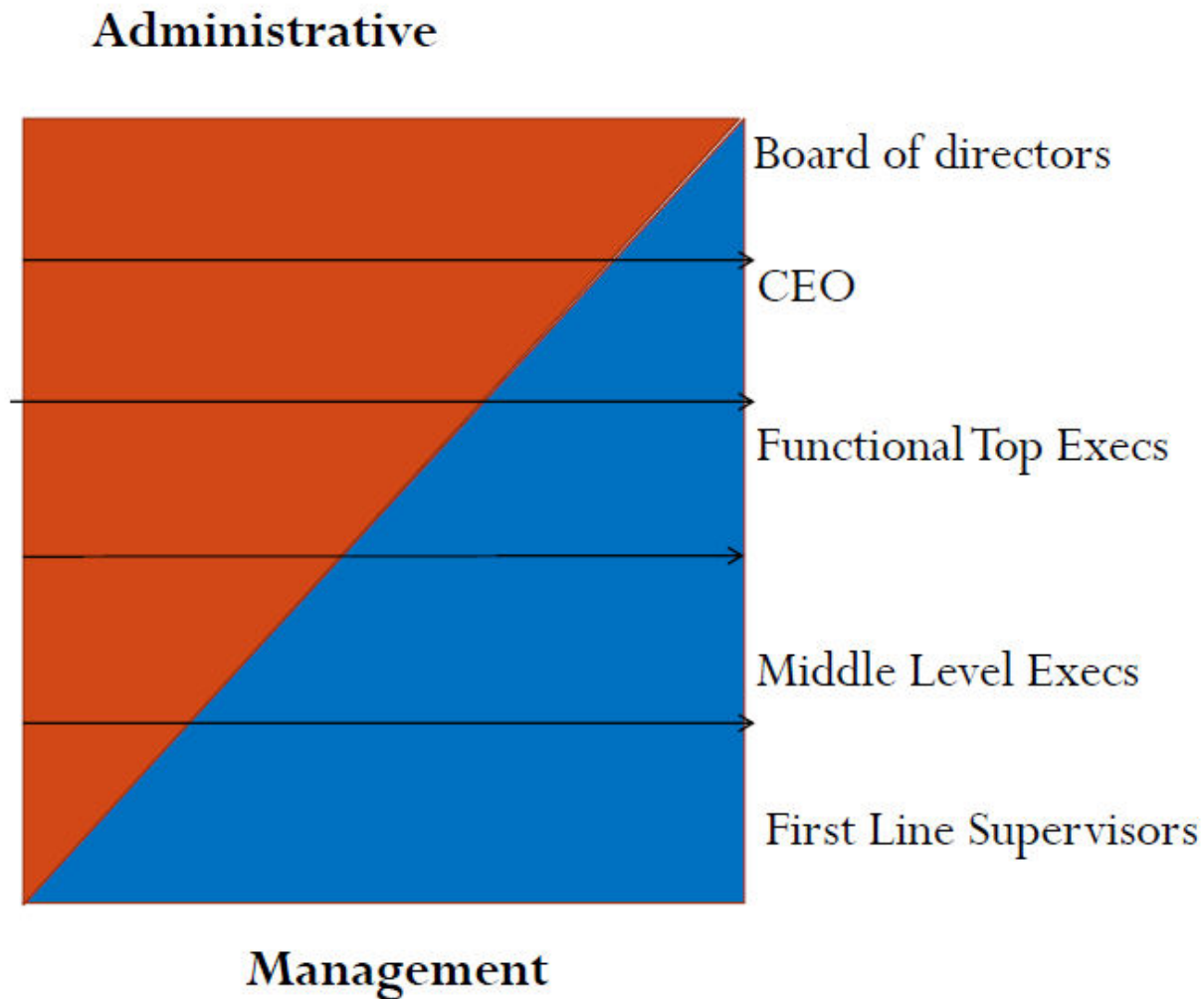
Relative Amount of Time That Managers Spend on the Four Managerial Functions



LEVELS OF MANAGEMENT



LEVELS OF MANAGEMENT



MANAGERIAL SKILLS

Managerial skills is the ability of a manager to make a smooth functioning team of people working under him. It involves obligation to make effective utilization of human and material resources. It requires sound judgment to handle complex situations.

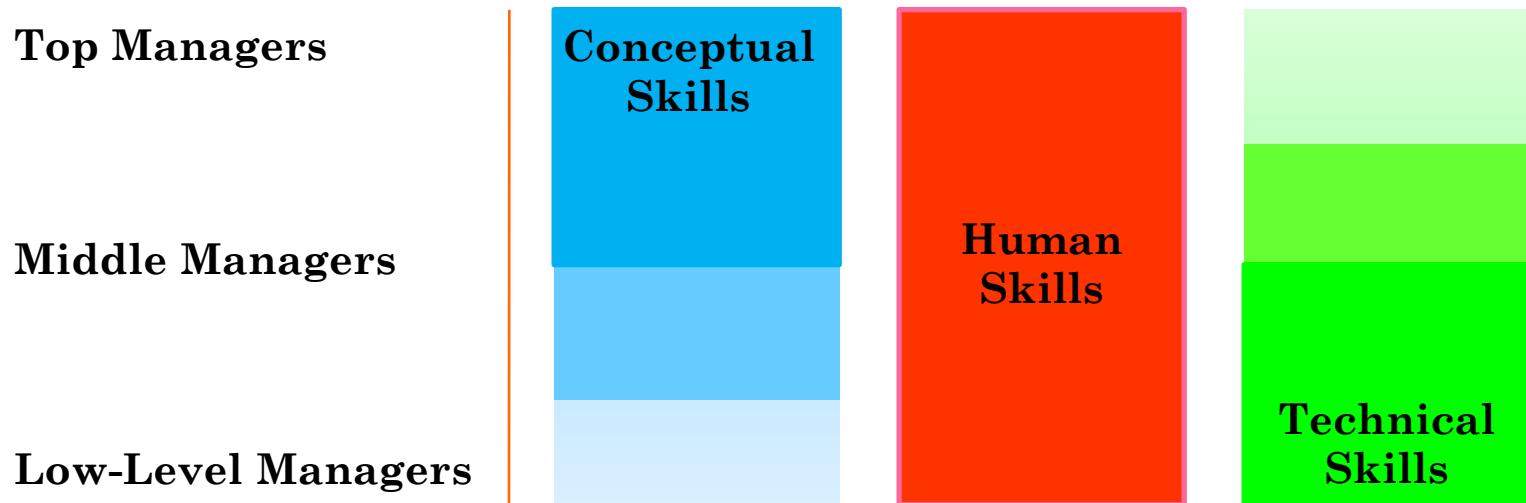
The skills required of a successful manager, whether he is working in a business organization, an educational institute or a hospital can be classified as under:

- a. *Technical Skills*
- b. *Conceptual Skills*
- c. *Human Relation Skills*

MANAGERIAL SKILLS

- ***Conceptual skills***: The ability to analyze and diagnose a situation and distinguish between cause and effect.
 - *Decision making skills*: ability to take timely and accurate decisions.
 - *Organizational skills*: ability to place right men for the right job.
- ***Human skills***: The ability to understand, alter, lead, and control the behavior of other individuals and groups.
 - *Communicating skills*: ability to pass on information.
 - *Motivating skills*: ability to inspire the subordinates.
 - *Leadership skills*: ability to inspire confidence and trust in the sub-ordinates.
- ***Technical skills***: Job-specific skills required to perform a particular type of work or occupation at a high level.
 - *ability to use methods, processes, tools, equipment, techniques and knowledge of a specialized field.*

SKILLS NEEDED AT DIFFERENT MANAGERIAL LEVELS



***Dark color = necessary to have**



PRINCIPAL OF MANAGEMENT (BY HENRI FAYOL)

○ *Division of Work*

- segregating work in the workforce amongst the worker will enhance the quality of the product.
- division of work improves the productivity, efficiency, accuracy and speed of the workers.

○ *Authority and Responsibility*

- Authority facilitates the management to work efficiently, and responsibility makes them responsible for the work done under their guidance or leadership.

○ *Discipline*

- Without discipline, nothing can be accomplished. It is the core value for any project or any management.

○ *Unity of Direction*

- Whoever is engaged in the same activity should have a unified goal.
- This means all the person working in a company should have one goal and motive which will make the work easier and achieve the set goal easily.

PRINCIPAL OF MANAGEMENT

○ *Unity of Command*

- This means an employee should have only one boss and follow his command.
- If an employee has to follow more than one boss, there begins a conflict of interest and can create confusion.

○ *Subordination of individual interest to general interest*

- This indicates a company should work unitedly towards the interest of a company rather than personal interest.

○ *Remuneration*

- This plays an important role in motivating the workers of a company. Remuneration can be monetary or non-monetary.

○ *Centralization*

- In any company, the management or any authority responsible for the decision-making process should be neutral.
- there should be a balance between the hierarchy and division of power.

○ *Scalar Chain*

- Hierarchy steps should be from the top to the lowest.
- This is necessary so that every employee knows their immediate senior also they should be able to contact any, if needed.

PRINCIPAL OF MANAGEMENT

○ *Order*

- A company should maintain a well-defined work order to have a favourable work culture.
- The positive atmosphere in the workplace will boost more positive productivity.

○ *Equity*

- All employees should be treated equally and respectfully.
- It's the responsibility of a manager that no employees face discrimination.

○ *Stability of tenure of Personnel*

- An employee delivers the best if they feel secure in their job.
- Duty of the management to offer job security to their employees.

○ *Initiative*

- The management should support and encourage the employees to take initiatives in an organization.

○ *Esprit de corps/Team Sprite*

- It is the responsibility of the management to motivate their employees and be supportive of each other regularly.
- Developing trust and mutual understanding will lead to a positive outcome and work environment.

SCOPE OF MANAGEMENT

- ***Production Management:*** Creation of Utilities by converting raw material into final product.
- ***Marketing Management:*** Involves distribution of product to the buyers: advertising, sales manager, market research
- ***Personnel Management:*** Effective use and control of man power: Personnel planning, selection & recruitment, training and development
- ***Financial Management:*** Deals with managerial activities related to procurement and utilization of fund for business: Budgeting, taxation
- ***IT Management:*** Deals with the use of IT to accelerate the business, use of services, communication: Network Management, Software management & Customization etc.

IMPORTANCE OF MANAGEMENT

- *Achievement of Group goals*
- *Optimum utilization of Resources*
- *Minimization of cost*
- *Survival and Growth*
- *Generation of Employment*
- *Development of the nation*

ADMINISTRATION VS MANAGEMENT

- Administration: Concerned with *policy formulation, obtaining finance, production & distribution.*
- *Thinking Process*
- Management: Concerned with *execution of policies set by administration and supervision of work*
- *Doing Process*

1.2 ORGANIZATION

Meaning and Concept of Organization

Characteristics of Organization

Principles of Organization

Formal and informal Organizations

Organization chart

Types of Organization-

Line, Line and staff

Functional and Matrix.

Authority and Responsibility and their interrelationships.

DEFINITION OF ORGANIZATIONS

- An organization is a group of people working together in a coordinated and structured fashion to achieve one or more goals. In organization
 - Human association
 - Achieve specific goals
 - Structured
 - Technology
 - Operate in dynamic environment
- Organizations use management to accomplish the work that is required to achieve the goals.



DEFINITION OF ORGANIZATIONS

- “Organization are planned units, deliberately structured for the purpose of attaining specific goals” By Amitai Etzioni
- “ Organization is a systematic arrangement of people brought together to accomplish some specific purpose” By Robbins and Delenzo
- “Organization is a group of people working together in structured and coordinated fashion to achieve a set of goals” By Ricky Griffin.



ORGANIZATIONS ROLE IN SOCIETY

- Organizations exist to allow accomplishment of work that could not be achieved by people alone.
- As long as the goals of an organization are appropriate, society will allow them to exist and they can contribute to society.



ORGANIZATIONS AND PEOPLE

- Organizations are strongly influenced by the people that form part of them.
- Organizations can take in part of the personality of the people within them and their attitudes, perceptions and behaviors affect how an organization will operate.



THE NATURE OF THE ORGANIZATIONAL ENVIRONMENT

- The *external environment* is everything outside an organization that might affect it.
- The *internal environment* consists of conditions and forces within the organization.



THE EXTERNAL ENVIRONMENT

- The general environment is the nonspecific dimensions and forces in its surroundings that might affect its activities.
- The task environment consists of specific organizations or groups that are likely to influence an organization.



THE INTERNAL ENVIRONMENT

- Board of Directors
- Employees
- Culture



CHARACTERISTICS OF ORGANIZATION

- Goal- orientation
- Collection of people
- Structure
- Technology
- Continuity
- environment



CHARACTERISTICS OF ORGANIZATION

- Goal orientation: profit, survival, growth, service, welfare, efficiency, leadership.
- Collection of people: Human association
- Structure: formal relationships and position of people in organization.
involves differentiation of activities and integration of efforts
 - differentiation is division of work
 - integration is coordination of individual efforts.



CHARACTERISTICS OF ORGANIZATION

- Technology: knowledge and skills required to perform activities.
- Continuity: stability, directly proportion to sustainability
- Environment: political, economic, socio-cultural and technological.



TYPES OF ORGANIZATION

- **Business:** Profit making organization. Eg. production company , service oriented organization (Bank)
- **Government:** For the welfare of people. Eg. Ministries, Police, Military etc.
- **Service:** Non profit organization. Eg. NGO, Health and Education organization etc.
- **Mutual Benefit:** For the benefit of member interest. Eg. Unions, Association of employers, NEA, Chamber of Commerce etc.
- **International:** Collective interest of member countries. Eg. SAARC, WHO etc.



PRINCIPLES OF ORGANIZATION

- **Principle of goal:** the organization should have specific goal and it is based on unity of goal so that all efforts can be concentrated on achieving the set goals- at minimum cost.
- **Division of Labor and Departmentalization**
 - *Division of labor* - Degree to which tasks are subdivided into separate jobs.
 - Specialization can lead to efficiency and increased performance, but if jobs become too specialized and boring, performance can decrease.
 - *Departmentalization* – Grouping of related activities into units.



PRINCIPLES OF ORGANIZATION

○ **Chain of Command**

- The line of authority from all level of managers to subordinates should be clearly defined and unbroken.

○ **Span of Management**

- It is the number of employees reporting to a manager. The number of employees reporting to one manager affects the number of levels of managers.
- With downsizing, the trend has clearly been to increase the span of management.



PRINCIPLES OF ORGANIZATION

○ Unity of Command

- Each subordinate should have only one boss and no person should report to more than one boss.

○ Delegation of Authority

- It should be adequate to accomplish expected results.

○ Parity of Authority and Responsibility

- Authority should be equal to responsibility.



PRINCIPLES OF ORGANIZATION

○ **Absoluteness of responsibility**

- Superiors can not escape responsibility for the activities of their subordinates.

○ **Balance**

- Balance in the structure regarding size span of management delegation of authority.

○ **Principle of seperation**

- Employees should not report to the department whose activity they are expected to evaluate.



PRINCIPLES OF ORGANIZATION

- **Flexibility**

The organization should be flexible enough to permit slight alternations and expansions whenever needed, due to changed circumstances.

- **Principle of leadership facilitation**

Organization should provide the facilities to develop the leadership.

- **Communication**

Not only executive should pass down information to the subordinates , there should be feedback i.e., replies should come from those who receive information. For better results it should have *free two-way communications*.

These principles are not rigid and they must be used as per local circumstances.



FORMAL AND INFORMAL ORGANIZATIONS

Formal Organizations

It is deliberately created by management for achieving the organization goals. It is a network of official authority responsibility relationships and communication among people working together to achieve common goals.

It is based on

- Division of work
- Division of authority
- Communication channel
- Superior-subordinates
- Formal policies, plans, rules, procedures
- Long life
- Eg. Government organization, Public organization



FORMAL AND INFORMAL ORGANIZATIONS

Informal organization

It is based on personal relationships rather than positional authority. It provides social satisfaction and may gain recognition. Here authority comes from knowledge, ability and experience. Eg. Social groups and clubs.

Characteristics

- Unplanned and spontaneous
- Results from human interactions
- Based on common interest
- No written policies, plans, procedures and rules
- Voluntary
- Quickly adopt
- Informal communication



FORMAL AND INFORMAL ORGANIZATIONS

Reasons for emergence of Informal organization

- Physically Proximity

Members who are physically located together affiliate with each other

- Similarity

Members with similar attitudes, interests and work related needs are attracted to one another

- Social Comparison

Members join informal groups for social comparison

- Interaction

Interactions, activities and sentiments lead to formation of informal groups



FORMAL AND INFORMAL ORGANIZATIONS

- **Origin:** Formal organization is created deliberately and consciously by management. Informal organization emerges spontaneously on account of socio psychological forces operating at the work place.
- **Purpose:** Formal organization is created for achieving the legitimate objective of the organization . Informal organization is created by the members of the organization for their social and psychological satisfaction.
- **Size:** Formal group may be quite large in size. Informal groups tend to be small.



FORMAL AND INFORMAL ORGANIZATIONS

- **Nature of group:** Formal groups are stable and may continue for a very long period of time. Informal groups were quite unstable in nature.
- **Number of group:** Generally the number of informal group is greater than the number of the formal group.
- **Authority:** Formal organization is bound together by a hierarchical structure. In Informal organization all members are equal.

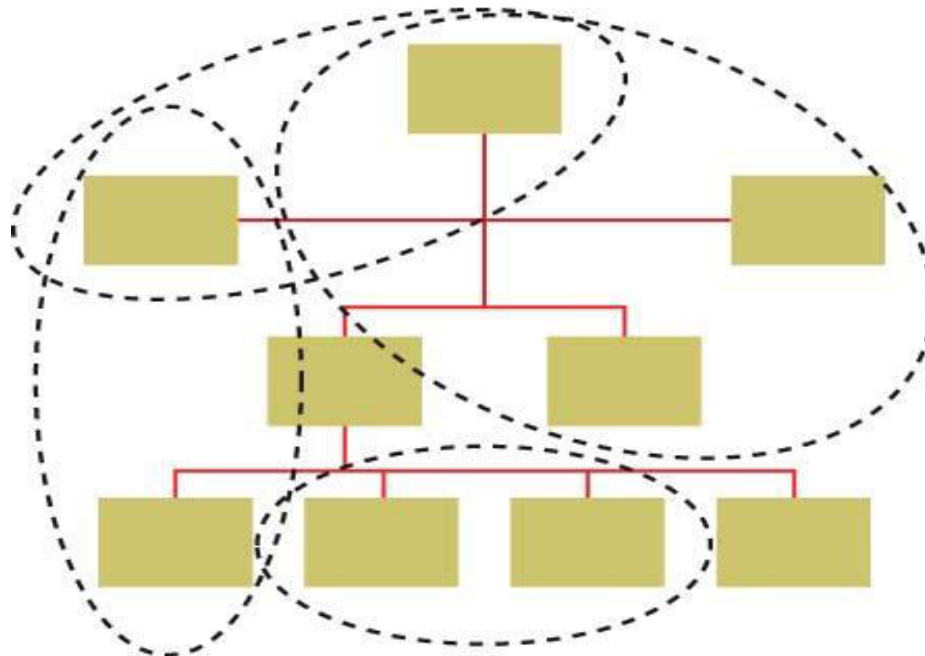


INFORMAL ORGANIZATION STRUCTURE

○ Informal Structure

- Unofficial but important working relationships between members

**Informal structures create a
“shadow” organization**



Shadow organizations can have both a positive and a negative affect on the organization's performance. In the case of weak management, the shadow organization can be what holds the business together.

IMPORTANCE OF MANAGEMENT IN TECHNOLOGY-DRIVEN ENVIRONMENTS

- Management ensures alignment between technological innovation and business goals.
- Facilitates efficient resource allocation and decision-making.
- Promotes collaboration across multidisciplinary teams.
- Addresses challenges in rapidly evolving markets through strategic planning.



IMPORTANCE OF MANAGEMENT IN TECHNOLOGY-DRIVEN ENVIRONMENTS

- **Risk Management:** Identifying, assessing, and mitigating technological and operational risks effectively.
- **Change Management:** Facilitating smooth transitions during the adoption of new technologies and processes.
- **Employee Development:** Investing in upskilling and training to keep the workforce updated with the latest technological advancements.
- **Customer-Centric Approach:** Aligning technological innovations with customer needs and feedback for enhanced satisfaction.
- **Sustainability Focus:** Driving eco-friendly practices in technology use and development to meet global sustainability goals.
- **Performance Monitoring:** Using advanced analytics and metrics to track and optimize operational efficiency and outcomes.



ENGINEERING FUNCTIONS IN ORGANIZATIONS

- **Product Development:**
 - Conceptualizing, designing, and launching new products.
- **Operations:**
 - Ensuring smooth production processes and efficiency.
- **IT Systems:**
 - Managing infrastructure, data, and technological solutions.
- **Quality Assurance:**
 - Maintaining product and service standards.
- **Other Functions:**
 - Research & Development, Maintenance, Supply Chain Management.



ROLES AND RESPONSIBILITIES OF AN ENGINEERING MANAGER

- **Leadership:**

- Guiding teams to achieve project goals and objectives.

- **Resource Management:**

- Allocating budgets, tools, and human resources effectively.

- **Communication:**

- Acting as a bridge between technical teams and stakeholders.

- **Decision-Making:**

- Evaluating options and implementing strategies for success.

- **Continuous Improvement:**

- Promoting innovation and adapting to technological advancements.



ROLES AND RESPONSIBILITIES OF AN ENGINEERING MANAGER

- **Strategic Planning:**

Developing long-term plans and aligning engineering activities with organizational goals.

- **Team Development:**

Mentoring and training team members to enhance their skills and career growth.

- **Risk Assessment:**

Identifying potential risks in projects and implementing mitigation strategies.

- **Budget Management:**

Ensuring that projects are executed within the allocated budget while maintaining quality standards.



ROLES AND RESPONSIBILITIES OF AN ENGINEERING MANAGER

- **Stakeholder Engagement:**

Building relationships with clients, vendors, and other key stakeholders to ensure project success.

- **Compliance Oversight:**

Ensuring adherence to industry regulations, safety standards, and ethical practices.

- **Technology Integration:**

Overseeing the adoption and implementation of new technologies to improve efficiency and innovation.

- **Performance Evaluation:**

Monitoring team and individual performances and providing constructive feedback.



Thank You